

Chapter 30: Homeostasis—Osmotic Regulation, Excretion, and Temperature Regulation

Homeostasis: Concept and Importance

- **Homeostasis** is the maintenance of a stable internal environment in the body.
- The concept was developed by **Claude Bernard** and named by **Walter B. Cannon**.



- **Walter B. Cannon** developed the concept of **homeostasis**:
 - Emphasized the importance of internal stability for survival.
 - Highlighted the role of physiological processes in restoring balance after disturbance.
 - His work laid the foundation for understanding feedback regulation in animals.
- Homeostasis is achieved through the coordinated activities of the **circulatory**, **nervous**, and **endocrine** systems, as well as organs like **kidneys**, **lungs**, **digestive tract**, and **integument**.
- **Negative feedback** regulation returns variables to a setpoint when deviations occur.

Water and Osmotic Regulation

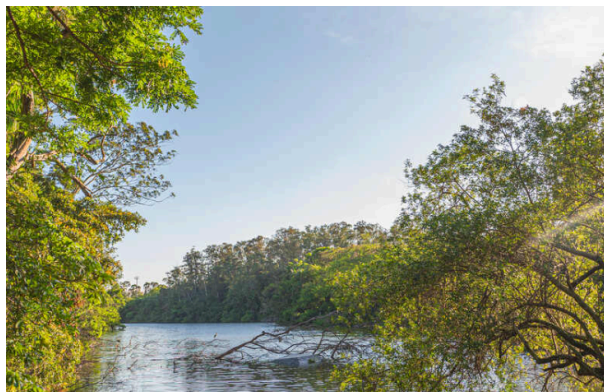
- **Osmotic regulation** maintains internal solute concentrations for optimal cellular function.
- **Osmotic conformers** (e.g., most marine invertebrates) have body fluids in equilibrium with seawater.
- **Osmotic regulators** actively control internal osmotic pressure, maintaining stability despite environmental changes.
- **Osmotic conformity** means internal osmotic pressure matches the environment; **osmotic regulation** means maintaining a different internal osmotic pressure.



- **Figure 30.2** illustrates two osmoregulatory strategies in crabs:
 - **Spider crab** is an **osmotic conformer**: its body fluid concentration changes with seawater salinity.
 - **Shore crab** is a **hyperosmotic regulator**: maintains higher internal salt concentration than dilute seawater.
 - **Tolerance limits** are shown: each species survives only within a specific salinity range.
- **Stenohaline** animals tolerate only a narrow range of salinity; **euryhaline** animals tolerate a wide range.
- **Hyperosmotic regulators** (e.g., brackish-water shore crabs) maintain body fluids more concentrated than the environment; **hypoosmotic regulators** (e.g., marine bony fishes) maintain body fluids less concentrated than the environment.
- **Active transport** in gills compensates for salt loss in dilute environments.

Osmoregulation in Freshwater and Marine Animals

- **Freshwater animals** are **hyperosmotic regulators**; they excrete excess water and actively absorb salts.



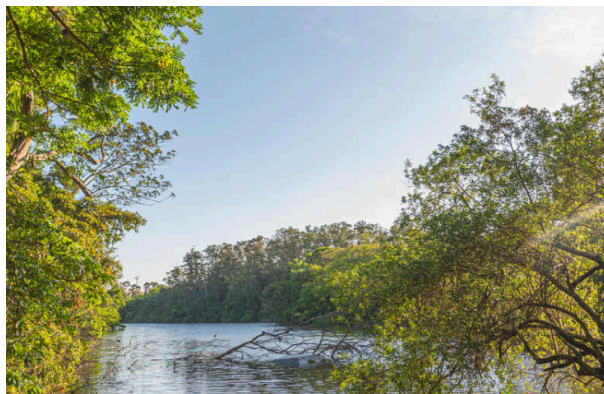
- **Figure 30.3** compares osmoregulation in freshwater and marine bony fishes:
 - **Freshwater fish**: absorb salts through gills, excrete large amounts of dilute urine, prevent water overload.
 - **Marine fish**: drink seawater, excrete excess salts via gills and kidneys, produce small amounts of concentrated urine.
 - Shows adaptations for maintaining internal osmotic balance in different environments.
- **Amphibians** absorb salts through their skin.



- **Figure 30.4** demonstrates osmoregulation in amphibians:
 - Water enters through permeable skin.
 - Kidneys excrete excess water as dilute urine.
 - Skin and kidneys actively reabsorb sodium chloride to maintain salt balance.
- **Marine bony fishes** are **hypoosmotic regulators**; they drink seawater and excrete excess salts via gills and kidneys.
- **Sharks and rays** maintain osmotic balance by retaining **urea** and **TMAO** in their blood.

Water Balance in Terrestrial Animals

- Terrestrial animals lose water by evaporation, excretion, and feces, and replace it by drinking, food, and metabolic water.
- **Kangaroo rats** survive in deserts without drinking water, relying on metabolic water and producing concentrated urine and dry feces.



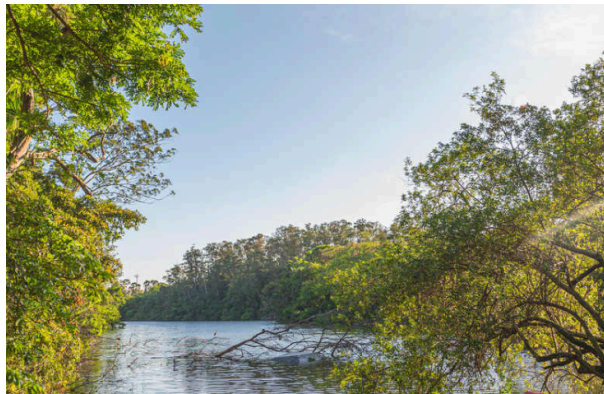
- **Table 30.1** compares water balance in humans and kangaroo rats:
 - **Kangaroo rats** obtain most water from metabolism, produce very concentrated urine and dry feces, and minimize evaporative loss.
 - **Humans** rely more on drinking water and lose more water through urine and evaporation.
 - Highlights adaptations for desert survival without drinking.
- **Salt glands** in marine birds and reptiles excrete excess salt.

Invertebrate Excretory Structures

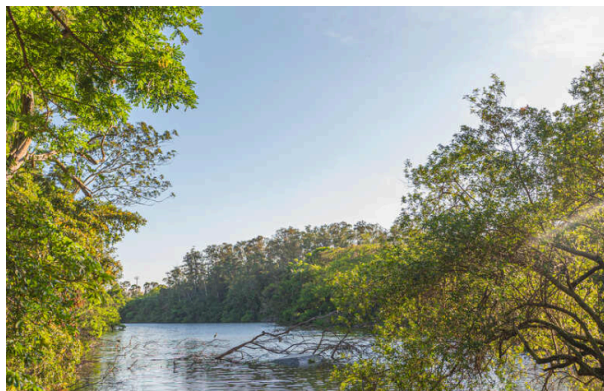
- **Contractile vacuoles** in protozoa and sponges expel excess water.
- **Flame-cell system** (protonephridia) in flatworms collects and expels wastes.



- **Figure 30.5** shows the flame-cell (protonephridial) system in flatworms:
 - **Flame cells** collect excess fluid from body tissues.
 - Fluid moves through branched ducts, valuable substances are reabsorbed.
 - Wastes are expelled through excretory pores on the body surface.
- **Metanephridia** in annelids and molluscs are open at both ends and reclaim valuable materials.

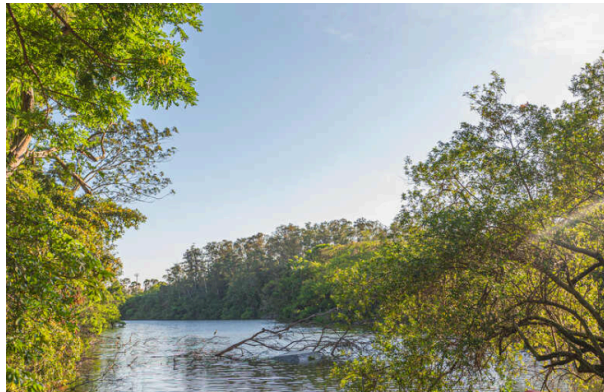


- **Figure 30.6** illustrates the metanephridial system in earthworms:
 - Each segment contains a pair of nephridia with ciliated funnels (nephrostomes).
 - Coelomic fluid enters the nephrostome, passes through the tubule, and valuable materials are reabsorbed into capillaries.
 - Wastes are excreted through nephridiopores.
- **Antennal glands** in crustaceans filter blood and form urine.



- **Figure 30.7** shows the structure of crustacean antennal glands:
 - Blood is filtered in the end sac to form ultrafiltrate.

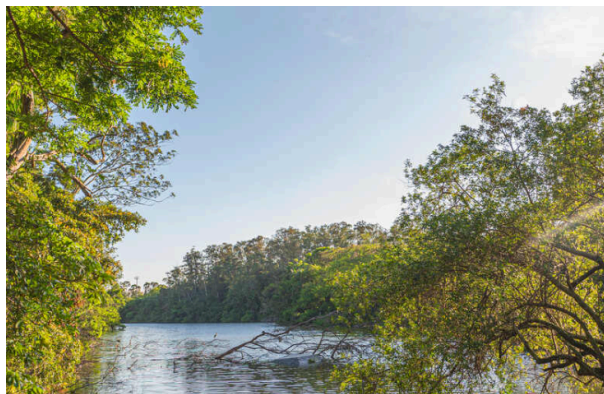
- Tubule modifies filtrate by selective reabsorption and secretion.
- Final urine is stored in the bladder and excreted.
- **Malpighian tubules** in insects secrete wastes and reabsorb water and salts.



- **Figure 30.8** explains the Malpighian tubule system in insects:
 - Tubules secrete ions and wastes from hemolymph into the lumen.
 - Water and solutes follow by osmosis.
 - Rectal glands reabsorb water and valuable ions, leaving uric acid and wastes for excretion.

Vertebrate Kidney: Structure and Function

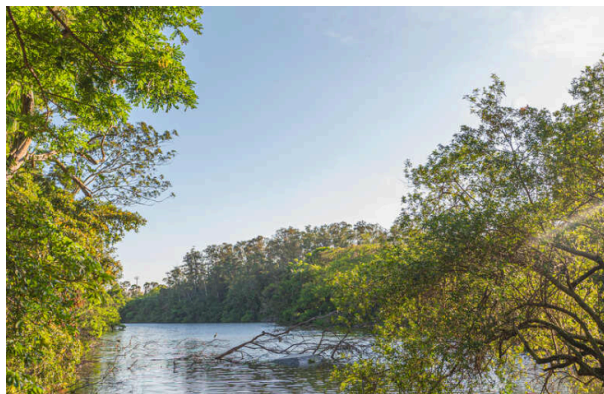
- The **archinephros** is the ancestral vertebrate kidney; modern kidneys develop through **pronephros**, **mesonephros**, and **metanephros** stages.



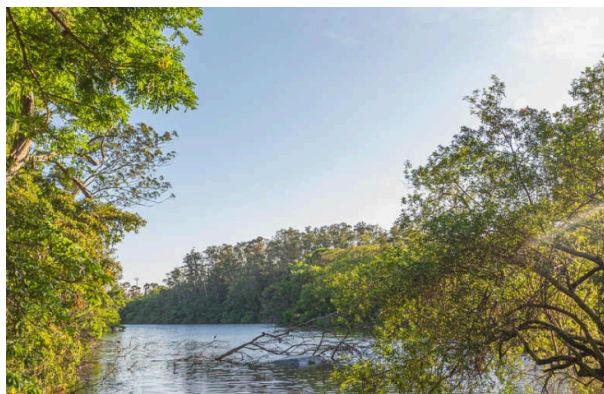
- **Figure 30.9** compares kidney development in vertebrates:
 - **Pronephros** is the earliest, transient kidney stage.
 - **Mesonephros** and **metanephros** develop later; metanephros forms the adult kidney in amniotes.
 - Shows association between excretory and reproductive ducts.
- The **nephron** is the functional unit of the kidney, consisting of a **glomerulus** and a **renal tubule**.



- **Figure 30.10** details the structure of the mammalian urinary system:
 - Shows kidney anatomy, nephron structure, and blood flow.
 - Illustrates the path of filtrate from Bowman's capsule through the tubule to the collecting duct.
 - Highlights the integration of filtration, reabsorption, and secretion.
- Blood is filtered in the glomerulus; filtrate passes through the tubule, where reabsorption and secretion occur.

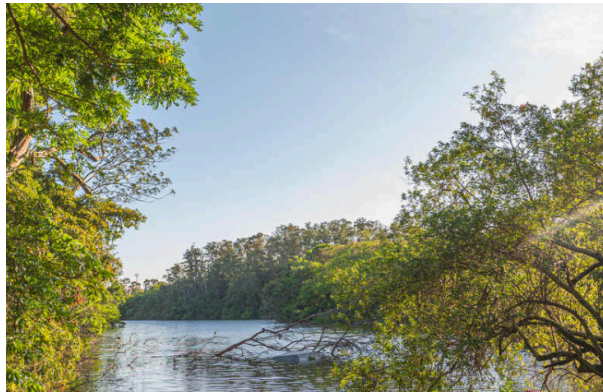


- **Figure 30.11** is a scanning electron micrograph of kidney microcirculation:
 - Shows glomeruli and associated blood vessels.
 - Demonstrates the extensive capillary network for filtration and exchange.
- The **renal corpuscle** filters plasma, excluding proteins and blood cells.



- **Figure 30.12** illustrates the renal corpuscle and filtration barrier:
 - Water, electrolytes, and small molecules pass through the glomerular membrane.
 - Plasma proteins and blood cells are retained in the blood.

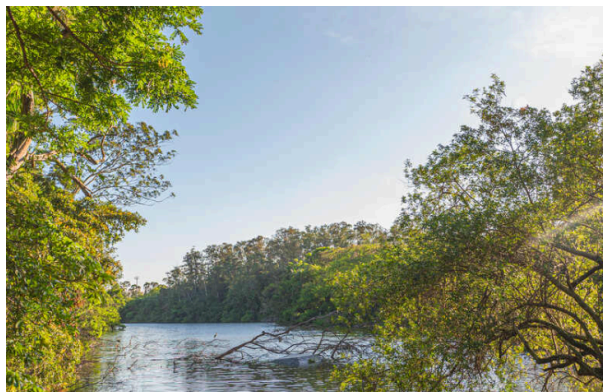
- The filtrate is almost protein-free.
- **Glucose** is reabsorbed up to a transport maximum; excess appears in urine (as in **diabetes mellitus**).



- **Figure 30.13** explains glucose reabsorption in the nephron:
 - At low concentrations, all glucose is reabsorbed.
 - When the transport maximum is reached, excess glucose appears in urine.
 - Demonstrates the mechanism underlying glycosuria in diabetes mellitus.
- **Aldosterone** and **antidiuretic hormone (ADH)** regulate sodium and water reabsorption.
- **Filtration** in the glomerulus produces a protein-free filtrate from blood plasma.
- **Tubular reabsorption** returns valuable substances (e.g., glucose, ions, water) from the filtrate to the blood.
- **Tubular secretion** moves substances (e.g., H^+ , K^+ , drugs) from blood into the filtrate for excretion.

Urine Concentration and Countercurrent Mechanisms

- The **loop of Henle** and **collecting duct** create an osmotic gradient in the kidney medulla, enabling urine concentration.



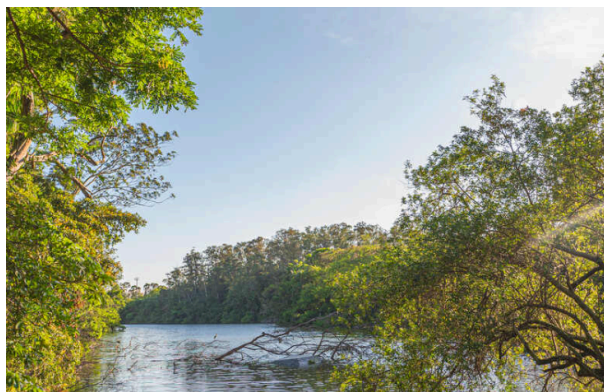
- **Figure 30.14** shows the mechanism of urine concentration in mammals:
 - Sodium and chloride are pumped out of the ascending limb of the loop of Henle.
 - Water is withdrawn from the descending limb.
 - Urea reabsorbed from the collecting duct increases medullary osmolarity.
- Osmotic concentration increases from cortex to medulla.



- **Figure 30.15** illustrates the osmotic concentration gradient in the kidney:
 - Osmolarity increases from cortex to medulla, peaking at the papilla.
 - This gradient enables the kidney to produce concentrated urine.
- **Countercurrent multiplication** and **urea recycling** contribute to high medullary osmolarity.
- **Countercurrent multiplication** is the process by which the loop of Henle creates a gradient of increasing osmolarity in the medulla, allowing urine concentration.
- **Urea recycling** increases medullary osmolarity by allowing urea to diffuse from the collecting duct into the medulla and back into the loop of Henle.
- The ability to concentrate urine depends on the length of the loop of Henle.

Temperature Regulation in Animals

- **Q₁₀** is the factor by which a biological rate increases for a 10°C rise in temperature.
- **Q₁₀** is calculated as the ratio of a rate at temperature $T + 10$ to the rate at temperature T .
- **Ectotherms** rely on environmental heat; **endotherms** generate metabolic heat.
- **Poikilothermic** animals have variable body temperature; **homeothermic** animals maintain constant temperature.
- **Ectothermy** means body temperature is determined by the environment; **endothermy** means body temperature is regulated by internal metabolic heat.
- **Behavioral thermoregulation** in ectotherms includes basking and seeking shade.



- **Figure 30.16** demonstrates behavioral thermoregulation in lizards:
 - Lizards bask in the sun to absorb heat in the morning.
 - Seek shade or burrow to avoid midday heat.
 - Adjust body orientation and position to regulate temperature throughout the day.

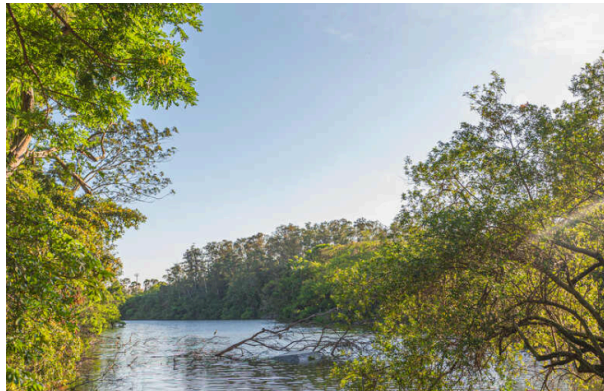
- **Endotherms** balance heat production and loss; heat is produced by metabolism and lost by radiation, conduction, convection, and evaporation.



- **Figure 30.17** shows heat exchange in a large mammal:
 - Heat is gained by solar radiation and lost by evaporation, conduction, convection, and longwave radiation.
 - The balance of these processes determines body temperature on a warm day.

Adaptations to Hot and Cold Environments

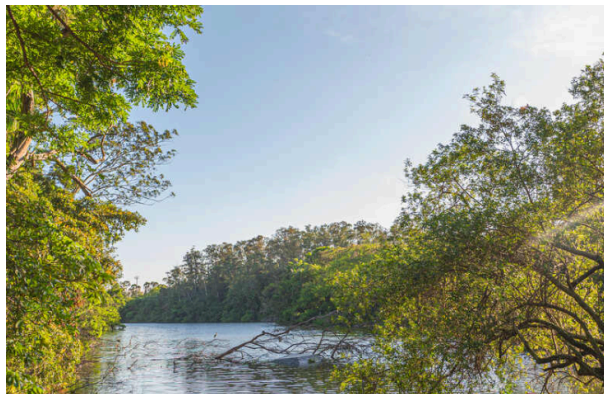
- Desert mammals (e.g., eland, camels) use insulation, variable body temperature, and concentrated urine to conserve water and avoid overheating.



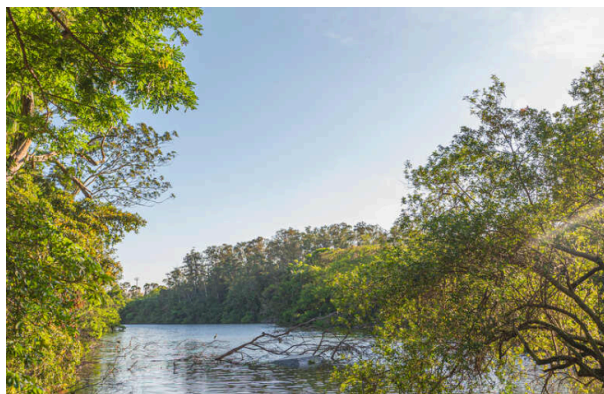
- **Figure 30.18** summarizes adaptations of the eland for hot, arid environments:
 - Light fur reflects sunlight, reducing heat absorption.
 - Insulating coat and fat storage in a hump minimize heat gain.
 - Variable body temperature reduces water loss by evaporation.
 - Produces concentrated urine and dry feces to conserve water.
- Arctic mammals use **countercurrent heat exchange** to conserve heat in extremities.



- **Figure 30.19** explains countercurrent heat exchange in arctic mammals:
 - Warm arterial blood transfers heat to cold venous blood returning from extremities.
 - Minimizes heat loss from legs and feet in cold environments.
 - Allows extremities to function at low temperatures without excessive heat loss.
- Small mammals and birds may enter **torpor** or **hibernation** to conserve energy.



- **Figure 30.20** shows torpor in hummingbirds:
 - Body temperature and oxygen consumption drop significantly during periods of food shortage or at night.
 - Torpor conserves energy and reduces metabolic demands.



- **Figure 30.21** depicts hibernation in a woodchuck:
 - Body temperature and metabolic rate decrease drastically during hibernation.
 - Breathing and heart rate slow to minimal levels.
 - Hibernation allows survival during prolonged cold and food scarcity.
- **Estivation** is dormancy during hot or dry periods.

- **Salt glands** are found in marine birds, turtles, and lizards, excreting highly concentrated salt solutions to maintain osmotic balance.

Key Formulas and Terms

- **Osmolarity**: $\text{osmolarity} = n \times C$, where n is the number of particles per molecule, C is molar concentration.
- **Q_10**: $Q_{10} = \frac{\text{rate}_{(T+10)}}{\text{rate}_T}$
- **Countercurrent multiplication**: mechanism in the loop of Henle for concentrating urine.
- **Negative feedback**: regulatory mechanism returning variables to a setpoint.